

The parity-separation constant of level- Q divisor weights tends to zero

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Abstract

For an integer N , put $Q = \lfloor \sqrt{N} \rfloor$ and let q_1 be the least prime exceeding Q . Split the squarefree Q -friable integers in $(\lfloor N/q_1 \rfloor, N]$ into an even-parity set $\mathcal{E}$ and an odd-parity set $\mathcal{O}$, according to their number of prime factors. Let $c_{\mathcal{E}}$ and $c_{\mathcal{O}}$ be their divisor-incidence count vectors, indexed by squarefree $d \leq Q$, and let K_Q be the cone generated by the incidence columns of $\mathcal{E}$. We prove that

$$\kappa(N) := \frac{\text{dist}_2(c_{\mathcal{O}}, K_Q)}{\|c_{\mathcal{E}} - c_{\mathcal{O}}\|_2} \ll_A (\log Q)^{-A} \quad \text{for every fixed } A > 0.$$

Thus nonnegative reweighting of the even-parity columns improves on uniform weighting by a factor smaller than every fixed negative power of $\log Q$. By conic duality, $\kappa(N)$ also bounds the normalised negative mean of a level- Q divisor weight on $\mathcal{O}$ when that weight is nonnegative on $\mathcal{E}$. The construction groups columns by their small-prime part and matches the odd-parity contribution of each retained class on smooth rows. We prove a uniform comparison of the two parity counts within these classes. Since $N < q_1^2$, every integer under consideration has at most one prime factor above Q ; the resulting exact decomposition reduces the signed estimate to quantitative prime number and Möbius estimates. We also give a two-term quantitative bound and certified finite-size computations. The implied constants are effective, but no numerical threshold is evaluated.

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1 Introduction

1.1 Definitions

Throughout, N is an integer and

$$Q = \lfloor \sqrt{N} \rfloor, \quad L = \log Q, \quad q_1 = \min\{p \text{ prime} : p > Q\}, \quad \text{thr} = \lfloor N/q_1 \rfloor, \quad (1)$$

with $\log$ the natural logarithm. For $n > 1$ let $P^-(n)$, $P^+(n)$ be the least and greatest prime factors of n , with $P^-(1) = \infty$ and $P^+(1) = 1$; $\omega(n)$ counts distinct prime factors, and μ is the Möbius function. An integer n is y -smooth (or y -friable) if $P^+(n) \leq y$, and y -rough if $P^-(n) > y$. Define the *band*

$$\mathcal{B} = \{n \in (\text{thr}, N] : \mu^2(n) = 1, P^+(n) \leq Q\}, \quad (2)$$

split by the parity of ω into

$$\mathcal{E} = \{n \in \mathcal{B} : \omega(n) \text{ even}\}, \quad \mathcal{O} = \{n \in \mathcal{B} : \omega(n) \text{ odd}\}, \quad R = |\mathcal{O}|. \quad (3)$$

Index coordinates by the *rows*, the squarefree $d \leq Q$. With $c_{\mathcal{E}}(d) = \#\{n \in \mathcal{E} : d \mid n\}$ and $c_{\mathcal{O}}(d) = \#\{n \in \mathcal{O} : d \mid n\}$, and assuming $R > 0$, set

$$t_d = \frac{c_{\mathcal{O}}(d)}{R}, \quad p_d = \frac{c_{\mathcal{E}}(d) - c_{\mathcal{O}}(d)}{R} = \frac{1}{R} \sum_{n \in \mathcal{B}, d \mid n} \mu(n), \quad (4)$$

the second equality because $\mu(n) = (-1)^{\omega(n)}$ on squarefree n . For $\ell \in \mathcal{E}$ let $A_{\ell} = (\mathbf{1}_{d \mid \ell})_d$ be its divisor-incidence column, let A be the matrix with these columns, put

$$K_Q = \text{cone}\{A_{\ell} : \ell \in \mathcal{E}\}, \quad (5)$$

and define

$$\kappa(N) = \frac{\text{dist}_2(t, K_Q)}{\|p\|_2}, \quad (6)$$

both norms being Euclidean on the row space. The vector p is the *parity defect* of the band: p_1 is the normalised Möbius sum over $\mathcal{B}$, and p_d its analogue along the row d .

Uniform weighting and normalisation. The homogeneity of distance to a cone gives

$$\kappa(N) = \frac{\text{dist}_2(c_{\mathcal{O}}, K_Q)}{\|c_{\mathcal{E}} - c_{\mathcal{O}}\|_2}. \quad (7)$$

We use this expression on the same domain as (6): $R > 0$ and $p \neq 0$. Proposition 8 establishes both conditions for all sufficiently large N . The uniform weights $x_{\ell}^{(0)} = 1/R$ give $Ax^{(0)} = t + p$, so

$$0 \leq \kappa(N) \leq 1. \quad (8)$$

The numerator of (7) is the smallest error obtainable by nonnegative reweighting; the denominator is the error from uniform weighting.

The level Q alone does not determine κ : the integers $Q^2 \leq N < (Q+1)^2$ may give different bands. For example, at $Q = 5$ the odd part is $\{5\}$ when $N = 25$, whose incidence vector does not belong to the cone generated by the columns of 6, 10, 15. When $N = 30$, the odd part is $\{5, 30\}$ and $t = \frac{1}{2}(A_{10} + A_{15}) \in K_Q$. Thus $\kappa(25) > 0 = \kappa(30)$, which explains the notation $\kappa(N)$.

1.2 Dual formulation

Let $\lambda = (\lambda_d)_{d \leq Q}$, $\mu^2(d)=1$ be a weight of *level* Q . Set $\lambda_d = 0$ for nonsquarefree $d \leq Q$, and write $\theta_{\lambda}(n) = \sum_{d \mid n, d \leq Q} \lambda_d$ for its divisor sum. Let $\mathbb{E}_{\mathcal{O}}$ denote the mean over $\mathcal{O}$.

Proposition 1. *Let N be such that $R > 0$. Then*

$$\text{dist}_2(t, K_Q) = \max\{-\mathbb{E}_{\mathcal{O}}\theta_{\lambda} : \|\lambda\|_2 \leq 1, \theta_{\lambda}(\ell) \geq 0 \text{ for all } \ell \in \mathcal{E}\}.$$

If $t \in K_Q$ the maximum is 0, attained at $\lambda = 0$. If $t \notin K_Q$ it is positive and attained at a unique λ , which satisfies $\|\lambda\|_2 = 1$.

Proposition 1, proved in §3, identifies $\kappa(N)\|p\|_2$ with the largest negative mean on $\mathcal{O}$ of a divisor weight that is nonnegative on $\mathcal{E}$, subject to a unit Euclidean norm bound on its coefficients. We call $\kappa(N)$ the *parity-separation constant* for level- Q divisor weights in this ℓ^2 normalisation. The sign constraint is imposed only on $\mathcal{E}$, and the norm is taken on the coefficients λ_d , rather than in Selberg's quadratic form. Changing the norm changes the constant, although the divisor-sum sign constraints remain the same.

1.3 Relation to sieve weights

Divisor-sum sign conditions also occur in sieve theory, but the condition $\theta_\lambda \geq 0$ on $\mathcal{E}$ lacks the normalisation and majorisation requirements of a classical upper-bound sieve weight. Here the primal problem is nonnegative least squares, and the dual in Proposition 1 is a second-order cone problem. The classical parity obstruction concerns what a sieve can establish without distinguishing the parity of ω [10, ch. 16]; the present result concerns the specific incidence cone K_Q .

The approximation problem involves more than comparing the cardinalities of $\mathcal{E}$ and $\mathcal{O}$. It asks for simultaneous approximation of their divisor-incidence counts over all squarefree $d \leq Q$. The number of rows grows with N , and the error is measured relative to the parity defect, not to the total number of integers in the band.

1.4 Results

Theorem 2. *For every fixed $A > 0$,*

$$\kappa(N) \ll_A (\log Q)^{-A}.$$

In particular $\kappa(N) \rightarrow 0$: by (8) the optimal nonnegative reweighting beats the uniform baseline by more than any fixed power of $\log Q$.

The following quantitative estimate implies Theorem 2 by taking $\beta = 2A + 4$.

Theorem 3. *Let $\beta \geq 4$ be fixed and put $Y = L^\beta$. For every $\varepsilon > 0$ there is an $N_0(\beta, \varepsilon)$ such that for every integer $N \geq N_0(\beta, \varepsilon)$, with $Q = \lfloor \sqrt{N} \rfloor$ and $L = \log Q$,*

$$\kappa(N) \leq \left(\frac{4}{\sqrt{\beta}} \sqrt{\frac{15}{\pi^2}} + \varepsilon \right) \frac{L^{2-\beta/2}}{\sqrt{\log L}} + (4\sqrt{2e^{e-1}} + \varepsilon) L^2 \exp\left(-\frac{L}{4\beta \log L}\right).$$

All constants and thresholds here may depend on β , which is fixed before N grows; no statement is claimed uniformly in β .

For fixed β , the second term is smaller than every power of $1/L$, so the first term determines the rate. The choice $\beta = 6$ gives the following numerical example.

Corollary 4. *Put*

$$c_1 = \frac{4}{\sqrt{6}} \sqrt{\frac{15}{\pi^2}} = 2.0131684\dots, \quad c_2 = 4\sqrt{2e^{e-1}} = 13.3565762\dots$$

For every $\varepsilon > 0$ and all sufficiently large N ,

$$\kappa(N) \leq \frac{c_1 + \varepsilon}{L\sqrt{\log L}} + (c_2 + \varepsilon) L^2 \exp\left(-\frac{L}{24 \log L}\right) \quad \text{and} \quad \kappa(N) \leq \frac{2.01317 + \varepsilon}{\log Q \sqrt{\log \log Q}}.$$

The decimal 2.01317 is rounded upward from c_1 . The coefficients are not asserted to be optimal.

Scope. Theorems 2 and 3 give upper bounds for this incidence-cone problem. They do not imply that $\kappa(N) > 0$ or give a matching lower bound, and they do not resolve the classical parity obstruction. The certified finite-size results in §11 are separate computations and are not used in the analytic proof.

1.5 Method

The proof exhibits a feasible point of K_Q close to t . Fix a parameter $\beta \geq 4$ and put

$$Y = L^\beta, \quad a_Y(n) = \prod_{p|n, p \leq Y} p, \quad (9)$$

and group the band by its *small-prime part* a_Y : for squarefree Y -smooth a let B_a and R_a be the number of elements of $\mathcal{E}$ and of $\mathcal{O}$ with $a_Y = a$, and set

$$x_\ell = \frac{R_{a_Y(\ell)}}{R B_{a_Y(\ell)}} \quad \text{if } a_Y(\ell) \leq \sqrt{Q} \text{ and } B_{a_Y(\ell)} > 0; \quad x_\ell = 0 \quad \text{otherwise.} \quad (10)$$

Then $x \geq 0$, so $Ax \in K_Q$ and $\text{dist}_2(t, K_Q) \leq \|t - Ax\|_2$. Under the hypothesis below, the weights match the odd contribution of every retained class on each Y -smooth row. Only the discarded odd classes contribute to the error there (Lemma 10); a Rankin estimate makes their total mass smaller than every power of $1/L$ (Lemma 14). On rows with a prime factor above Y , the bounds $t_d, (Ax)_d = O(1/d)$ give an ℓ^2 error of order $(Y \log Y)^{-1/2}$ (Lemmas 12 and 13). Theorem 9 combines these estimates under the following parity-comparison hypothesis.

Hypothesis 5 (PC). There is an absolute C with $R_a \leq C B_a$ for every squarefree Y -smooth $a \leq \sqrt{Q}$.

Hypothesis 5 says that after fixing the small-prime part, the two ω -parities of the rough cofactor occur in bounded ratio. We prove it with $C = 1 + O((\log L)^{-2})$ (Theorem 21). Writing $n = ab$ with $a = a_Y(n)$, the cofactor b runs over squarefree integers with all prime factors in $(Y, Q]$ and $b \leq N/a \in [Q^{3/2}, N]$, and the parity split of such b is governed by

$$U(X) = \#\{b \leq X : \mu^2(b) = 1, Y < P^-(b), P^+(b) \leq Q\}, \quad M_{Y,Q}(X) = \sum_{\substack{b \leq X \\ Y < P^-(b), P^+(b) \leq Q}} \mu(b). \quad (11)$$

Hypothesis 5 follows from $U(X) \gg XV(Y)$ together with $|M_{Y,Q}(X)| = o(U(X))$, where $V(y) = \prod_{p \leq y} (1 - 1/p)$. The unsigned lower bound follows by decomposing according to the prime factor above Q and applying a uniform estimate for rough integers (Lemmas 15 and 16).

For the signed estimate we use the following elementary fact: since $Q = \lfloor \sqrt{N} \rfloor$,

$$N < (Q + 1)^2 \leq q_1^2,$$

so an integer $n \leq N$ has at most one prime factor exceeding Q , and that prime divides n to the first power. Factoring out this prime leaves a cofactor below q_1 , so its prime factors are automatically at most Q . The usual decomposition by the largest prime factor therefore gives the exact identity (Lemma 18)

$$M_{Y,Q}(X) = M_Y(X) + \sum_{Q < p \leq X} M_Y(X/p), \quad M_Y(z) = \sum_{b \leq z, P^-(b) > Y} \mu(b).$$

The inner sums have no additional restriction on the largest prime factor. Quantitative prime number and Möbius estimates, stated as (P) and (L) in §3.1, then give the signed estimate (Lemma 19 and Theorem 20).

To pass from an absolute error to a bound for κ , we also need a lower bound for $\|p\|_2$. Section 7 proves $\|p\|_2 \geq |p_1| = (1 + o(1))N/(4RL^2)$ by the same decomposition at $d = 1$ and an exact integral transform, with main term from $\sum_{n \geq 1} \mu(n) \log n/n = -1$. That section also records the density $R \gg N$. These are special cases of known friable asymptotics (§2.2); the proofs are included for the range needed here.

2 Related work

2.1 Programming formulations of sieve weights

Brady's thesis [4] formulates combinatorial sieve weights as a linear program with pointwise divisor-sum sign constraints. It studies duality (§§3.1, 4.1, 7.1), a model optimisation problem (§4.2), numerical optima at finite level (§8.1), and Selberg's parity example with two-term asymptotics (§8.2). His later paper [5, §2, Prop. 1] develops a semidefinite formulation and proves co-NP-hardness for verifying divisor-sum sign constraints in its input representation. In the present problem, the sign condition is imposed only on the even-parity part of an explicitly specified band, and the cone is generated by scalar incidence columns. These constraints differ from the partition cones of positive semidefinite matrices in [5]; no complexity claim is made here.

Related programming formulations appear in Heath-Brown [11, §4] and Ford–Maynard [7, Thms. 7.3–7.4]. Rigorous numerical estimates for sieve constants appear in [13, §7.1], [3, §§5.1–5.2], and [8, §§7–8], the last using proved approximation-error bounds.

2.2 Möbius sums over friable and over rough integers

The counting estimates in §5 and the normalisation in §7 are related to the following results on friable and rough integers.

- $U(X)$ in (11) is the squarefree analogue of the two-cutoff count $\Theta(x, y, z) = \#\{n \leq x : \text{all prime factors in } (z, y]\}$. That count was first studied by Friedlander [9], who for fixed values of the two ratios of logarithms obtained an asymptotic whose factor is defined by a system of differential-difference equations; sharper estimates are due to Saias [15], continued in [16] (for this ordering see [12, §7]). All of them count unrestricted integers. For the unrestricted analogue, specialising the modulus in [18, (1.4)] to the primorial through Y compares the sifted friable count with $\Psi(x, y)$, whose size is given by [12, Thm. 1.1]; passing to squarefree integers requires a further adjustment. Lemma 16 is a weak consequence.
- Lemma 17 is a weak form of [12, Cor. 1.3], which gives $\Psi(x, y) = x u^{-(1+o(1))u}$ as y and $u = \log x / \log y$ tend to infinity, uniformly for $u \leq y^{1-\varepsilon}$, a range that strictly contains the one used here.
- Lemma 15 is the Buchstab asymptotic restricted to $u \rightarrow \infty$; see [19, ch. III.6].
- Alladi [1] studies Möbius sums over integers free of small prime factors; [2] treats sums over friable integers. These are the respective objects of Lemma 19 and §7. The convolution and estimates used here are proved below.
- The asymptotic $P_1 = -N/(\log N)^2 + O(N/(\log N)^3)$ of §7 is such a friable Möbius sum, at $u = \log N / \log Q \rightarrow 2$. For it in the form used here see de la Bretèche and Tenenbaum [6, Cor. 1.3, (1.23)]: specialising to μ and taking $y = \sqrt{N} + \frac{1}{2}$ puts u below 2, which satisfies the one-sided exclusion condition there, and gives leading term $x \varpi'(u) / \log^2 y$ with $\varpi'(u) = -1/u^2$, i.e. $-N/\log^2 N$; changing the cutoff from y to Q affects at most one prime and costs $O(\sqrt{N})$, as does the lower band edge. Our §7 is an elementary derivation in the single case needed, not a new asymptotic.

2.3 Contribution

The contribution is the class reweighting (10) and its application to the relative distance (6). The large-prime decomposition and the analytic estimates are standard in origin; we give proofs in the required ranges and make no separate novelty claim for Theorem 20.

3 Notation, inputs, and elementary lemmas

In addition to (1)–(9) we write

$$\theta = \frac{1}{\log Y}, \quad Z_0 = \exp\left(\frac{L}{(\log L)^3}\right), \quad S = \sum_{p \leq Y} \frac{1}{p}, \quad V(y) = \prod_{p \leq y} \left(1 - \frac{1}{p}\right), \quad (12)$$

and, for $z \geq 1$,

$$\Phi(z) = \#\{n \leq z : P^-(n) > Y\}, \quad M_Y(z) = \sum_{b \leq z, P^-(b) > Y} \mu(b), \quad M(x) = \sum_{n \leq x} \mu(n), \quad (13)$$

with $n = 1$ counted in Φ ; also $\Psi(z, y) = \#\{n \leq z : P^+(n) \leq y\}$ and $Q_2(x) = \sum_{n \leq x} \mu^2(n)$, and γ is Euler's constant; $M(x) = M(\lfloor x \rfloor)$ for real x .

Asymptotic convention. Throughout, β is fixed independently of N . Unless stated otherwise, constants in O , o , and $\ll$ may depend on β , and on ε when it occurs. Subscripts emphasise dependence rather than asserting additional uniformity. By Mertens' theorems, $S \asymp \log \log L$ and $V(Y) \sim e^{-\gamma}/(\beta \log L)$.

Standing assumption. From §4 onward, N is assumed large enough that $R > 0$, $p \neq 0$ and

$$2 \leq Y < Q, \quad (14)$$

which is possible because $Y = (\log Q)^\beta$. In particular, (14) supplies the cutoff condition used in Lemma 18.

3.1 Classical inputs

Each is used only in the form stated.

- (M1) $\sum_{p \leq x} (\log p)/p = \log x + O(1)$; explicitly $< \log x$ for $x > 1$ [14, (3.24)].
- (M2) $\sum_{p \leq x} 1/p = \log \log x + O(1)$; and for $x \geq y \geq 2$, $\sum_{y < p \leq x} 1/p = \log(\log x / \log y) + O(1/\log y)$.
- (M3) $V(y) = e^{-\gamma}(\log y)^{-1}(1 + O(1/\log y))$.
- (C) Chebyshev: $\pi(x) \ll x/\log x$; explicitly $\pi(y) \leq 1.26 y/\log y$ for $y > 1$, a rounding of the bound $\pi(y) < 1.25506 y/\log y$ of [14, (3.6)].
- (P) There are absolute effective $C_\pi, c_\pi > 0$ with $|\pi(y) - \text{li}(y)| \leq C_\pi y \exp(-c_\pi \sqrt{\log y})$ for $y \geq 3$.
- (L) There are absolute effective $C_M, c_M > 0$ with $|M(x)| \leq C_M x \exp(-c_M (\log x)^{1/4})$ for real $x \geq 2$.

(M1)–(M3) and (C) are Mertens' and Chebyshev's theorems. Inputs (P) and (L) are both consequences, by standard contour integration, of the de la Vallée Poussin zero-free region [20, Thm. 3.8] together with the bounds for ζ'/ζ and $1/\zeta$ on it [20, Thm. 3.11] and truncated Perron inversion [20, Lem. 3.12]; both are quoted here in weakened shapes. For explicit constants in (P), Theorem 2 of [21] gives

$$|\pi(x) - \text{li}(x)| \leq 0.2795 x (\log x)^{-3/4} \exp\left(-\sqrt{\frac{\log x}{6.455}}\right) \quad (x \geq 229),$$

whose logarithmic factor is at most 1 for $x \geq e$ and whose exponential constant is $1/\sqrt{6.455} = 0.3935\dots$; enlarging C_π over the finite range $3 \leq x < 229$ yields (P) with $c_\pi = 0.39$. Nothing

below depends on the value of c_π , which enters only through $e^{-c_\pi\sqrt{L}}$ being smaller than every power of $1/L$. We also use

$$Q_2(x) = \frac{6}{\pi^2}x + O(\sqrt{x}), \quad (15)$$

which follows from $Q_2(x) = \sum_{d \leq \sqrt{x}} \mu(d) \lfloor x/d^2 \rfloor = x \sum_{d \leq \sqrt{x}} \mu(d) d^{-2} + O(\sqrt{x})$ and $\sum_{d > \sqrt{x}} d^{-2} \ll x^{-1/2}$.

Lemma 6. $\sum_{n \geq 1} \frac{\mu(n)}{n} = 0$ and $\sum_{n \geq 1} \frac{\mu(n) \log n}{n} = -1$.

Proof. By partial summation and (L), $\sum_{k > T} \mu(k)/k = -M(T)/T + \int_T^\infty M(v)v^{-2}dv \rightarrow 0$ as $T \rightarrow \infty$, and likewise with $\log k$ inserted; so both series converge. For $\sigma > 1$ we have $\sum_n \mu(n)n^{-\sigma} = \zeta(\sigma)^{-1}$ and, differentiating, $\sum_n \mu(n)(\log n)n^{-\sigma} = \zeta'(\sigma)/\zeta(\sigma)^2$. Since a Dirichlet series converging at $s = 1$ is continuous from the right there (Abel's theorem for Dirichlet series), the two sums equal $\lim_{\sigma \rightarrow 1+} \zeta(\sigma)^{-1} = 0$ and $\lim_{\sigma \rightarrow 1+} \zeta'(\sigma)/\zeta(\sigma)^2 = -1$, the latter from $\zeta(\sigma) = (\sigma - 1)^{-1} + \gamma + O(\sigma - 1)$ and $\zeta'(\sigma) = -(\sigma - 1)^{-2} + O(1)$. $\square$

Lemma 7. For every integer $N \geq 4$:

- (i) $Q^2 \leq N < (Q+1)^2 \leq q_1^2$; consequently every $n \leq N$ has at most one prime factor exceeding Q , and that prime divides n exactly once;
- (ii) $Q/2 - 1 \leq \text{thr} \leq Q$;
- (iii) $2L \leq \log N < 2L + 2/Q$, so $L/\log N \leq 1/2$.

Proof. (i) $Q = \lfloor \sqrt{N} \rfloor$ gives $Q+1 > \sqrt{N}$, so $N < (Q+1)^2$; and $q_1 \geq Q+1$ because q_1 is an integer exceeding Q . Two primes exceeding Q (equal or not) dividing $n \leq N$ would force $n \geq q_1^2 > N$. (ii) $N < (Q+1)^2 \leq q_1(Q+1)$ gives $N/q_1 < Q+1$, so $\text{thr} \leq Q$; and $q_1 \leq 2Q$ by Bertrand gives $\text{thr} \geq N/(2Q) - 1 \geq Q/2 - 1$. (iii) $Q^2 \leq N < (Q+1)^2$ and $\log(Q+1) < L + 1/Q$. $\square$

Part (i) is why Theorem 2 concerns every large N and not a subsequence: it replaces the hypothesis $N = Q^2$ in the large-prime decompositions of Lemmas 16, 18 and 22.

Proof of Proposition 1. K_Q is finitely generated, hence a closed convex cone, so with $K_Q^\circ = \{w : \langle w, v \rangle \leq 0 \ \forall v \in K_Q\}$ Moreau's decomposition (see e.g. [17, Thm. 1.1]) gives

$$\text{dist}_2(t, K_Q) = \max\{\langle t, w \rangle : w \in K_Q^\circ, \|w\|_2 \leq 1\}.$$

Put $\lambda = -w$; then $\langle w, A_\ell \rangle = \theta_w(\ell) \leq 0$ is the condition $\theta_\lambda(\ell) \geq 0$, and by (4)

$$\langle t, w \rangle = -\sum_d t_d \lambda_d = -\frac{1}{R} \sum_d c_{\mathcal{O}}(d) \lambda_d = -\frac{1}{R} \sum_{n \in \mathcal{O}} \theta_\lambda(n) = -\mathbb{E}_{\mathcal{O}} \theta_\lambda,$$

which is the displayed equality. If $t \in K_Q$ the distance is 0 and $w = 0$ attains it. If $t \notin K_Q$ the distance is positive and is attained at $w = (t - \Pi_{K_Q} t) / \|t - \Pi_{K_Q} t\|_2$. For uniqueness, suppose $w_1 \neq w_2$ both lie in K_Q° with $\|w_i\|_2 \leq 1$ and $\langle t, w_i \rangle = D > 0$. Their midpoint w lies in K_Q° , has $\langle t, w \rangle = D$, and has $\|w\|_2 < 1$ by strict convexity of the Euclidean ball; then $w/\|w\|_2 \in K_Q^\circ$ is feasible with value $D/\|w\|_2 > D$, a contradiction. The same rescaling shows $\|w\|_2 = 1$ at the maximum. $\square$

The following normalisation is proved in §7.

Proposition 8. Put $P_1 = \sum_{n \in \mathcal{B}} \mu(n)$, a quantity defined without dividing by R . There are absolute effective constants such that, for $\log N$ large,

$$P_1 = -\frac{N}{(\log N)^2} \left(1 + O\left(\frac{1}{\log N}\right) \right).$$

Consequently $P_1 < 0$ and $|\mathcal{B}| = \frac{6}{\pi^2}(1 - \log 2)N + O(N/L)$, so $R \geq \frac{1}{2}|\mathcal{B}| \geq 0.09N > 0$; hence $N/R = O(1)$, indeed $N/R \rightarrow \pi^2/(3(1 - \log 2)) = 10.7213228\dots$, and $p_1 = P_1/R$ is well defined and satisfies, for every $\varepsilon \in (0, 1)$ and all $L \geq L_0(\varepsilon)$,

$$\|p\|_2 \geq |p_1| \geq (1 - \varepsilon) \frac{N}{R} \cdot \frac{1}{4L^2} > 0.$$

4 The construction and the reduction

Group the band by small-prime part: for squarefree Y -smooth a put $\mathcal{E}(a) = \{\ell \in \mathcal{E} : a_Y(\ell) = a\}$, $B_a = |\mathcal{E}(a)|$, $\mathcal{O}(a) = \{n \in \mathcal{O} : a_Y(n) = a\}$, $R_a = |\mathcal{O}(a)|$, so that $\mathcal{E} = \bigsqcup_a \mathcal{E}(a)$ and $\mathcal{O} = \bigsqcup_a \mathcal{O}(a)$. With x as in (10) set

$$e_d = t_d - (Ax)_d, \quad \text{so} \quad \text{dist}_2(t, K_Q) \leq \|e\|_2, \quad (16)$$

and

$$\delta = \frac{1}{R} \#\{n \in \mathcal{O} : a_Y(n) > \sqrt{Q}\}. \quad (17)$$

For a retained class $a \leq \sqrt{Q}$, Hypothesis 5 implies $B_a = 0 \Rightarrow R_a = 0$. Without this implication, a class with $B_a = 0 < R_a$ contributes an additional error in (18), since the weights (10) cannot reproduce its odd mass. Classes with $a > \sqrt{Q}$ are discarded regardless of their parity counts.

Theorem 9. Assume Hypothesis 5 with constant C . For every $\varepsilon > 0$ and all $L \geq L_0(\varepsilon)$,

$$\kappa(N) \leq (4 + \varepsilon)L^2 \left[\sqrt{\frac{15}{\pi^2}} \max(1, C) \left(\sum_{p > Y} p^{-2} \right)^{1/2} + \left(\frac{2\delta R}{N} \right)^{1/2} \right].$$

4.1 The Y -smooth rows

Lemma 10. Let $d \leq Q$ be squarefree with $P^+(d) \leq Y$. Then, unconditionally,

$$e_d = \frac{1}{R} \left[\sum_{a > \sqrt{Q}, d|a} R_a + \sum_{a \leq \sqrt{Q}, B_a=0, d|a} R_a \right], \quad (18)$$

and under Hypothesis 5 the second sum vanishes, so

$$e_d = \frac{1}{R} \#\{n \in \mathcal{O} : d \mid n, a_Y(n) > \sqrt{Q}\} \in [0, \min(\delta, t_d)]. \quad (19)$$

In particular, under Hypothesis 5, $e_1 = \delta$: on the row $d = 1$ the error is then exactly the discarded odd mass. Unconditionally, (18) at $d = 1$ adds to that the mass of the retained classes with $B_a = 0$.

Proof. Every prime factor of d is at most Y and d is squarefree, so for squarefree ℓ one has $d \mid \ell$ if and only if $d \mid a_Y(\ell)$. Hence $\#\{\ell \in \mathcal{E}(a) : d \mid \ell\} = B_a \mathbf{1}_{d|a}$ and, by (10),

$$(Ax)_d = \frac{1}{R} \sum_{a \leq \sqrt{Q}, B_a > 0, d|a} R_a, \quad t_d = \frac{1}{R} \sum_{a: d|a} R_a,$$

the second because $\mathcal{O}$ is partitioned by a_Y . Subtracting gives (18): the classes absent from $(Ax)_d$ are those with $a > \sqrt{Q}$ and those with $a \leq \sqrt{Q}$, $B_a = 0$. For a retained class $a \leq \sqrt{Q}$ with $B_a = 0$, Hypothesis 5 forces $R_a \leq C \cdot 0 = 0$, so every term of the second sum is zero and the sum vanishes (its index set need not be empty). For the range in (19): $e_d \geq 0$ termwise; $e_d \leq t_d$ since $(Ax)_d \geq 0$; and $e_d \leq \delta$ since the set counted lies in $\{n \in \mathcal{O} : a_Y(n) > \sqrt{Q}\}$. $\square$

Lemma 11. *Assume Hypothesis 5 and put $c_0 = N/R$. Then $\|e|_{P^+(d) \leq Y}\|_2 \leq \sqrt{2c_0\delta}$.*

Proof. If $\delta = 0$ then $e_d = 0$ on every Y -smooth row by (19); so assume $\delta > 0$. Since $c_{\mathcal{O}}(d) \leq \lfloor N/d \rfloor \leq N/d$ we have $t_d \leq c_0/d$, so $0 \leq e_d \leq f(d)$ with $f(w) = \min(\delta, c_0/w)$. As f^2 is nonincreasing on $(0, \infty)$, $f(d)^2 \leq \int_{d-1}^d f(w)^2 dw$, whence

$$\sum_{d \geq 1} f(d)^2 \leq \int_0^\infty \min\left(\delta^2, \frac{c_0^2}{w^2}\right) dw = \int_0^{c_0/\delta} \delta^2 dw + \int_{c_0/\delta}^\infty \frac{c_0^2}{w^2} dw = 2c_0\delta. \quad \square$$

4.2 The rows with a large prime factor

The next estimate is used here and in the counting lemmas.

Lemma 12. *Unconditionally,*

$$\sum_{\substack{d \text{ squarefree} \\ P^+(d) > Y}} d^{-2} \leq \frac{\zeta(2)}{\zeta(4)} \sum_{p > Y} p^{-2}, \quad \sqrt{\frac{\zeta(2)}{\zeta(4)}} = \sqrt{\frac{15}{\pi^2}} = 1.2328088 \dots,$$

and $\sum_{p > Y} p^{-2} = (1 + o(1))/(Y \log Y)$ as $Y \rightarrow \infty$.

Proof. Every squarefree d with a prime factor exceeding Y is pm with $p > Y$ prime and m squarefree, so

$$\sum_{\substack{d \text{ squarefree} \\ P^+(d) > Y}} d^{-2} \leq \left(\sum_{p > Y} p^{-2} \right) \sum_{m \text{ squarefree}} m^{-2} = \left(\sum_{p > Y} p^{-2} \right) \prod_p (1 + p^{-2}) = \frac{\zeta(2)}{\zeta(4)} \sum_{p > Y} p^{-2},$$

because $1 + p^{-2} = (1 - p^{-4})/(1 - p^{-2})$; and $\zeta(2)/\zeta(4) = 15/\pi^2$. For the last claim, partial summation with $\pi(t) = (1 + o(1))t/\log t$ gives $\sum_{p > Y} p^{-2} = -\pi(Y)/Y^2 + 2 \int_Y^\infty \pi(t)t^{-3} dt = (1 + o(1))(-1 + 2)/(Y \log Y)$. $\square$

Lemma 13. *Assume Hypothesis 5. Then*

$$\|e|_{P^+(d) > Y}\|_2 \leq \max(1, C) \frac{N}{R} \sqrt{\frac{\zeta(2)}{\zeta(4)}} \left(\sum_{p > Y} p^{-2} \right)^{1/2}.$$

Proof. Let $d \leq Q$ be squarefree with a prime factor exceeding Y . Then $t_d = c_{\mathcal{O}}(d)/R \leq (N/R)/d$; and by (10) with Hypothesis 5 one has $x_\ell \leq C/R$ for every ℓ , so $0 \leq (Ax)_d \leq (C/R) \#\{\ell \in \mathcal{E} : d \mid \ell\} = Cc_{\mathcal{E}}(d)/R \leq C(N/R)/d$. Hence $|e_d| \leq \max(1, C)(N/R)/d$. Summing the squares over these d and applying Lemma 12 gives the claim. $\square$

4.3 The discarded mass

Lemma 14. *With $\theta = 1/\log Y$ and $\Xi = e^{e-1} = 5.5749415 \dots$,*

$$\sum_{n \leq N} a_Y(n)^\theta \leq \Xi N, \quad \text{hence} \quad \delta \leq \Xi \frac{N}{R} \exp\left(-\frac{L}{2 \log Y}\right).$$

Proof. Since $a_Y(n)$ is squarefree and Y -smooth, for $\theta > 0$

$$a_Y(n)^\theta = \prod_{p|n, p \leq Y} (1 + (p^\theta - 1)) = \sum_{d|a_Y(n)} \prod_{p|d} (p^\theta - 1) = \sum_{\substack{d|n, \mu^2(d)=1 \\ P^+(d) \leq Y}} \prod_{p|d} (p^\theta - 1),$$

because $d | a_Y(n)$ is the same as $d | n$ with d squarefree and Y -smooth. All coefficients are nonnegative. Summing over $n \leq N$ and using $\lfloor N/d \rfloor \leq N/d$,

$$\sum_{n \leq N} a_Y(n)^\theta \leq N \prod_{p \leq Y} \left(1 + \frac{p^\theta - 1}{p}\right).$$

For $p \leq Y$ we have $0 \leq \theta \log p \leq 1$, and $e^v - 1 \leq (e - 1)v$ on $[0, 1]$ by convexity, so by (M1)

$$\sum_{p \leq Y} \frac{p^\theta - 1}{p} \leq (e - 1)\theta \sum_{p \leq Y} \frac{\log p}{p} \leq (e - 1)\theta \log Y = e - 1;$$

now use $\prod(1+u_p) \leq \exp(\sum u_p)$. For the second display, Markov's inequality on $\mathcal{O}$ with threshold $\sqrt{Q}$ gives $\delta \leq R^{-1}Q^{-\theta/2} \sum_{n \in \mathcal{O}} a_Y(n)^\theta \leq (N/R) \Xi Q^{-\theta/2}$, and $Q^{-\theta/2} = \exp(-L/(2 \log Y))$. $\square$

4.4 Proof of Theorem 9

By (16) and Lemmas 11 and 13,

$$\text{dist}_2(t, K_Q) \leq \sqrt{2(N/R)\delta} + \max(1, C) \frac{N}{R} \sqrt{\frac{15}{\pi^2}} \left(\sum_{p > Y} p^{-2} \right)^{1/2}.$$

By Proposition 8, $1/\|p\|_2 \leq (4 + \varepsilon)L^2(R/N)$ for $L \geq L_0(\varepsilon)$. Multiplying by this bound gives the result: the factors N/R and R/N cancel in the second term, while $(R/N)\sqrt{2(N/R)\delta} = \sqrt{2\delta R/N}$ in the first. $\square$

5 The counting lemmas

The relevant range is $X \in [Q^{3/2}, N]$, i.e. $\log X \in [1.5L, 2L + O(1/Q)]$ by Lemma 7(iii). A single $Z_0 = \exp(L/(\log L)^3)$ serves every lemma below.

Lemma 15. *Uniformly for $z \geq Z_0$, $\Phi(z) = zV(Y)(1 + o(1))$.*

Proof. Let $P = \prod_{p \leq Y} p$, $J = \lceil \log L \rceil$, and for $n \geq 1$ put $k = \omega(\gcd(n, P))$. Legendre's identity reads $\mathbf{1}_{P-(n) > Y} = \sum_{d|\gcd(n, P)} \mu(d)$, and

$$\sum_{j=0}^J (-1)^j \binom{k}{j} = (-1)^J \binom{k-1}{J} \quad (k \geq 1), \quad = 1 \quad (k = 0).$$

For $k \geq 1$, the absolute truncation error is $\binom{k-1}{J}$; for $k = 0$, it is zero. In either case it is at most $\binom{k}{J}$. Summing over $n \leq z$ and using $\sum_{n \leq z} \binom{\omega(\gcd(n, P))}{J} = \sum_{d|P, \omega(d)=J} \lfloor z/d \rfloor$,

$$\left| \Phi(z) - \sum_{d|P, \omega(d) \leq J} \mu(d) \lfloor z/d \rfloor \right| \leq z \sum_{d|P, \omega(d)=J} \frac{1}{d} \leq \frac{zS^J}{J!}, \quad (20)$$

the last step because the elementary symmetric function obeys $e_J(1/p : p \leq Y) \leq S^J/J!$. Replacing $\lfloor z/d \rfloor$ by z/d and completing the Möbius sum,

$$\sum_{d|P, \omega(d) \leq J} \mu(d) \lfloor z/d \rfloor = zV(Y) - z \sum_{d|P, \omega(d) > J} \frac{\mu(d)}{d} + O(\#\{d | P : \omega(d) \leq J\}), \quad (21)$$

$$\left| \sum_{d|P, \omega(d) > J} \frac{\mu(d)}{d} \right| \leq \sum_{j > J} \frac{S^j}{j!} \leq \frac{S^{J+1}}{(J+1)!} \left(1 - \frac{S}{J+2}\right)^{-1} \leq \frac{2S^{J+1}}{(J+1)!} \quad \text{if } S \leq \frac{J+2}{2}, \quad (22)$$

$$\#\{d | P : \omega(d) \leq J\} \leq (J+1)\pi(Y)^J \leq Y^J. \quad (23)$$

The side condition in (22) holds eventually because $S \asymp \log \log L$ and $J \asymp \log L$; the second inequality of (23) holds eventually because (C) gives $Y/\pi(Y) \gg \log Y$, so $(Y/\pi(Y))^J \geq J+1$. Divide the three errors by $zV(Y)$, which is $\asymp z/(\beta \log L)$ by (M3). Since $S = (1+o(1)) \log \log L$ and $J = \lfloor \log L \rfloor$,

$$\frac{S^J}{J!} \leq \left(\frac{eS}{J}\right)^J = \exp(-(\log L)(\log \log L - \log \log \log L - 1 + o(1))),$$

smaller than every fixed power of $1/L$. Thus the corresponding coefficients in (20) and (22) are $o(V(Y))$. For (23) we compare logarithms: $\log(Y^J) = \beta J \log L = \beta(\log L)^2(1+o(1))$, whereas

$$\log(zV(Y)) \geq \log(Z_0V(Y)) = \frac{L}{(\log L)^3} - \log(\beta \log L) - \gamma + o(1),$$

and $\beta(\log L)^2 = o(L/(\log L)^3)$ because $(\log L)^5 = o(L)$. All three errors are therefore $o(1)$ relative to $zV(Y)$, uniformly in $z \geq Z_0$. $\square$

Lemma 15 asks only for $z \geq Z_0$, where $u = \log z / \log Y \geq L/(\beta(\log L)^4) \rightarrow \infty$; this avoids the Buchstab transition. For comparison, with ϖ Buchstab's function one has, uniformly for $2 \leq Y \leq \sqrt{z}$,

$$\Phi(z) = \frac{z}{\log Y} \left\{ \varpi(u) + O\left(\frac{1}{\log Y}\right) \right\}$$

[19, ch. III.6], hence $\Phi(z) \sim \varpi(u)e^\gamma zV(Y)$ as $Y \rightarrow \infty$, uniformly in that range, which contains the range of Lemma 15 since $2\beta(\log L)^4 \leq L$ eventually. Here $\max_{2 \leq u \leq 3} \varpi(u)e^\gamma = 1.0101232\dots$, attained at $u = 2.7632228\dots$, and the maximum can be read off the delay equation: on $[2, 3]$ one has $u\varpi(u) = 1 + \log(u-1)$, so $u^2\varpi'(u) = u/(u-1) - 1 - \log(u-1)$, which written in $v = u-1$ is $1/v - \log v$, strictly decreasing on $[1, 2]$ from 1 to $-0.1931\dots$ and therefore zero exactly once; there $v \log v = 1$ and $\varpi(u) = 1/v = 0.5671432\dots$. The argument below uses only the limit $\varpi(u)e^\gamma \rightarrow 1$ as $u \rightarrow \infty$, which yields the constant $1 - \log 2$ in Lemma 16.

Lemma 16. *Uniformly for $X \in [Q^{3/2}, N]$,*

$$U(X) \geq (1 - \log 2 + o(1)) X V(Y), \quad 1 - \log 2 = 0.3068528\dots,$$

and in particular $U(X) \gg X/\log Y$.

Proof. Let $\Phi_Q(z) = \#\{n \leq z : Y < P^-(n), P^+(n) \leq Q\}$. If $n \leq X \leq N$ has $P^-(n) > Y$ and $P^+(n) > Q$, then by Lemma 7(i) the large prime p is unique and divides n once; writing $n = pm$, Lemma 7(i)–(ii) give $m \leq X/p \leq N/q_1 < q_1$, so every prime factor of m is smaller than q_1 , i.e. at most Q , so the upper cutoff on m is automatic, and $P^-(m) > Y$ or $m = 1$. The map $n \mapsto (p, m)$ is a bijection onto the pairs counted, so

$$\Phi(X) = \Phi_Q(X) + \sum_{Q < p \leq X} \Phi_Q(X/p), \quad \text{hence} \quad \Phi_Q(X) \geq \Phi(X) - \sum_{Q < p \leq X} \Phi(X/p). \quad (24)$$

Split the subtracted sum at $X/p = Z_0$, using Lemma 15 for $p \leq X/Z_0$ and $\Phi(z) \leq z$ beyond:

$$\sum_{Q < p \leq X} \Phi(X/p) \leq (1 + o(1))XV(Y) \sum_{Q < p \leq X} \frac{1}{p} + X \sum_{X/Z_0 < p \leq X} \frac{1}{p}.$$

By (M2) and Lemma 7(iii), $\sum_{Q < p \leq X} 1/p = \log(\log X / \log Q) + O(1/L) \leq \log 2 + o(1)$, uniformly for $X \leq N$. For the short block put $\tau_X = \log Z_0 / \log X = L / ((\log L)^3 \log X)$; then

$$\sum_{X/Z_0 < p \leq X} \frac{1}{p} = -\log(1 - \tau_X) + O(1/L) = \tau_X + O(\tau_X^2 + 1/L) \leq \left(\frac{2}{3} + o(1)\right)(\log L)^{-3}, \quad (25)$$

because $\log X \geq 1.5L$ gives $\tau_X \leq \frac{2}{3}(\log L)^{-3}$, and $1/L = o((\log L)^{-3})$. The display (25) is an inequality: $\tau_X(\log L)^3$ equals $2/3$ at $X = Q^{3/2}$ and $\frac{1}{2} + O(1/(QL))$ at $X = N$. Since $(\log L)^{-3} = o(V(Y))$ the short block is $o(XV(Y))$, and $\Phi(X) = (1 + o(1))XV(Y)$ by Lemma 15; hence $\Phi_Q(X) \geq (1 - \log 2 + o(1))XV(Y)$. Finally, deleting the non-squarefree b costs at most $\sum_{p > Y} X/p^2 \ll X/(Y \log Y) = o(XV(Y))$ by Lemma 12, which is unconditional. $\square$

Lemma 17. *Let $u = \log z / \log Y$. Uniformly for $Z_0 \leq z \leq N$, so that $u \geq L/(\beta(\log L)^4) \rightarrow \infty$,*

$$\Psi(z, Y) \leq z \exp\left(-\frac{1}{2}u \log u\right).$$

Proof. Rankin's trick: for $\alpha \in (0, 1]$ one has $\Psi(z, Y) \leq z^\alpha \prod_{p \leq Y} (1 - p^{-\alpha})^{-1}$, since $\sum_{P^+(n) \leq Y} (z/n)^\alpha \geq \Psi(z, Y)$ termwise. Take $\alpha = 1 - (\log u)/(\log Y)$, so that $z^\alpha = z \exp(-u \log u)$ exactly. Throughout the range $\log u = (1 + o(1)) \log L$ and $\log Y = \beta \log L$, so $1 - \alpha = (1 + o(1))/\beta$ and $\alpha \rightarrow 1 - 1/\beta \in [3/4, 1)$, uniformly. The bound $\alpha \geq 3/4$ also holds as a stated inequality for L beyond an absolute point: $z \leq N$ gives $\log z < 2L + 2/Q \leq 3L$, so $u = \log z / \log Y \leq 3L/(\beta \log L) \leq L$, whence $\log u \leq \log L$ and

$$\alpha = 1 - \frac{\log u}{\beta \log L} \geq 1 - \frac{1}{\beta} \geq \frac{3}{4}$$

since $\beta \geq 4$. Estimate $\sum_{p \leq Y} p^{-\alpha}$ by Abel summation:

$$\sum_{p \leq Y} p^{-\alpha} = Y^{-\alpha} \pi(Y) + \alpha \int_2^Y \pi(t) t^{-\alpha-1} dt.$$

Since $Y^{1-\alpha} = u$ and $(1 - \alpha) \log Y = \log u$, the boundary term is, by (P),

$$Y^{-\alpha} \pi(Y) = (1 + o(1)) \frac{Y^{1-\alpha}}{\log Y} = (1 + o(1))(1 - \alpha) \frac{u}{\log u} = \left(\frac{1}{\beta} + o(1)\right) \frac{u}{\log u},$$

so the boundary term contributes to the leading coefficient. In the integral, the part $t \leq \sqrt{Y}$ is at most $\int_2^{\sqrt{Y}} t^{-\alpha} dt \leq Y^{(1-\alpha)/2} / (1 - \alpha) = O(u^{1/2} \log L) = o(u / \log u)$; on $[\sqrt{Y}, Y]$, (P) gives $\pi(t) = (1 + o(1))t / \log t$ uniformly, and substituting $t = Y^s$, so that $t^{1-\alpha} = u^s$ and $dt/(t \log t) = ds/s$,

$$\alpha \int_{\sqrt{Y}}^Y \pi(t) t^{-\alpha-1} dt = (1 + o(1)) \alpha \int_{1/2}^1 u^s \frac{ds}{s} = (1 + o(1)) \alpha \frac{u}{\log u} = \left(1 - \frac{1}{\beta} + o(1)\right) \frac{u}{\log u}$$

by Laplace's method, since the integral is concentrated near $s = 1$. The two contributions sum to $(1 + o(1))u / \log u$. For the Euler product, peel off $p = 2$, whose factor contributes $-\log(1 - 2^{-\alpha}) = O(1)$ because $2^{-\alpha} \leq 2^{-3/4} = 0.5946 \dots < 1$; for $p \geq 3$ and $\alpha \geq 3/4$ one has $p^{-\alpha} \leq 3^{-3/4} = 0.4386 \dots < 1/2$, so $-\log(1 - v) \leq v + v^2$ applies on its stated range, and $\sum_{3 \leq p \leq Y} p^{-2\alpha} \leq \sum_p p^{-3/2} = O(1)$. Hence $\log \prod_{p \leq Y} (1 - p^{-\alpha})^{-1} = (1 + o(1))u / \log u + O(1)$, and $\log(\Psi(z, Y)/z) \leq -u \log u + (1 + o(1))u / \log u + O(1) \leq -\frac{1}{2}u \log u$, since $u / \log u = o(u \log u)$. $\square$

6 Proof of the parity hypothesis

6.1 The parity split

Fix a squarefree Y -smooth $a \leq \sqrt{Q}$. Every $n \in \mathcal{B}$ with $a_Y(n) = a$ is $n = ab$ with b squarefree, $Y < P^-(b)$, $P^+(b) \leq Q$ and $b \in (\text{thr}/a, N/a]$; coprimality of a and b is automatic from $P^-(b) > Y \geq P^+(a)$, and $\omega(n) = \omega(a) + \omega(b)$. With

$$U_a = U(N/a) - U(\text{thr}/a), \quad M_a = M_{Y,Q}(N/a) - M_{Y,Q}(\text{thr}/a)$$

in the notation (11), and $\mu(b) = (-1)^{\omega(b)}$ for squarefree b ,

$$\{B_a, R_a\} = \left\{ \frac{U_a + M_a}{2}, \frac{U_a - M_a}{2} \right\}, \quad \frac{R_a}{B_a} \leq \frac{U_a + |M_a|}{U_a - |M_a|} \quad \text{if } |M_a| < U_a. \quad (26)$$

The order of the two counts is determined by the parity of $\omega(a)$. A bound $|M_a| \leq \rho U_a$ with a fixed $\rho < 1$ would suffice for Hypothesis 5; we will prove $|M_a| = o(U_a)$.

6.2 The signed estimate

Lemma 18. *Assume (14). Then for every $X \leq N$,*

$$M_{Y,Q}(X) = M_Y(X) + \sum_{Q < p \leq X} M_Y(X/p).$$

Proof. Split $M_Y(X)$ according to whether $P^+(b) \leq Q$. If not, $P^+(b) = p > Q$; by Lemma 7(i) p is the only prime factor of b exceeding Q and divides b once, so $b = pb'$ with $b' = b/p \leq X/p \leq N/q_1 < q_1$, whence every prime factor of b' is at most Q , $p \nmid b'$, and $P^-(b') > Y$ or $b' = 1$. Conversely, let $Q < p \leq X$ and $b' \leq X/p$ with $P^-(b') > Y$, and put $b = pb'$. Since

$$b' \leq X/p \leq N/q_1 < q_1 \leq p,$$

we have $p \nmid b'$ and hence $\mu(pb') = -\mu(b')$. Also $P^-(b) = \min(p, P^-(b')) > Y$, since $p > Q > Y$ by (14). Thus b is counted by $M_Y(X)$ with $P^+(b) > Q$, and the correspondence is a bijection. It follows that $M_Y(X) = M_{Y,Q}(X) - \sum_{Q < p \leq X} M_Y(X/p)$. $\square$

The hypothesis $Y \leq Q$ is needed: without it the converse direction fails. At $N = 256$, $Q = 16$, $Y = (\log 16)^6 > 256$ and $X = 17$, both sides of the identity would read 1 and 2 respectively, the discrepancy being the pair $(p, b') = (17, 1)$, whose product 17 is not counted by $M_Y(17)$.

Lemma 19. *Uniformly for $Z_0 \leq z \leq N$ and for every fixed $A_0 > 0$, $|M_Y(z)| \ll_{A_0, \beta} z(\log L)^{-A_0}$.*

Proof. Every n factors uniquely as $n = mb$ with m Y -friable and $P^-(b) > Y$, and $\mu(n) = \mu(m)\mu(b)$; in Dirichlet series, $\sum_{P^-(b) > Y} \mu(b)b^{-s} = \prod_{p > Y} (1 - p^{-s}) = \zeta(s)^{-1} \prod_{p \leq Y} (1 - p^{-s})^{-1}$, so $M_Y(z) = \sum_{m \leq z, P^+(m) \leq Y} M(z/m)$. Split at $m \leq z/W$ with $W = \exp(\sqrt{\log z})$. For $m \leq z/W$, apply (L) at argument at least W and use $\sum_{m \text{ } Y\text{-friable}} 1/m = V(Y)^{-1} \asymp e^\gamma \log Y$ from (M3):

$$\sum_{m \leq z/W} |M(z/m)| \ll z \exp(-c_M(\log W)^{1/4}) V(Y)^{-1} \ll z(\log L) \exp(-c_M(\log z)^{1/8}).$$

For $m > z/W$ use $|M(z/m)| \leq z/m \leq W$ and Lemma 17: that block is at most $W\Psi(z, Y) \leq z \exp(\sqrt{\log z} - \frac{1}{2}u \log u)$. In the stated range $\log u = (1 + o(1)) \log L$, so

$$u \log u = \frac{\log z}{\beta \log L} (1 + o(1)) \log L = (1 + o(1)) \frac{\log z}{\beta},$$

and $\sqrt{\log z} = o(\log z)$; hence $\sqrt{\log z} - \frac{1}{2}u \log u \leq -(\log z)/(3\beta)$ for L large, uniformly. Since $\log z \geq L/(\log L)^3$, both blocks are $\ll z \exp(-c'L^{1/9})$ with c' depending on β , and hence are $o(z(\log L)^{-A_0})$ for every fixed A_0 . $\square$

Theorem 20. *Uniformly for $X \in [Q^{3/2}, N]$,*

$$|M_{Y,Q}(X)| \leq \left(\frac{2}{3} + o(1)\right) X(\log L)^{-3}.$$

Proof. Apply Lemma 18 and split the prime sum at $X/p = Z_0$. First $|M_Y(X)| \ll X(\log L)^{-4}$ by Lemma 19, as $X \geq Q^{3/2} \gg Z_0$. Next, for $Q < p \leq X/Z_0$, Lemma 19 with $A_0 = 4$ gives $|M_Y(X/p)| \ll (X/p)(\log L)^{-4}$, while by (M2) with Lemma 7(iii) one has $\sum_{Q < p \leq X} 1/p = \log(\log X / \log Q) + O(1/L)$ uniformly, which since $X \leq N$ is at most $\log 2 + o(1)$, the limit at $X = Q^{3/2}$ being $\log(3/2)$; only the resulting $O(1)$ is used, so this block is $\ll X(\log L)^{-4}$. Finally, for $X/Z_0 < p \leq X$ use $|M_Y(z)| \leq z$ and (25), which gives at most $(\frac{2}{3} + o(1))X(\log L)^{-3}$. $\square$

6.3 Uniform parity comparison

Theorem 21. *For all L beyond a point depending only on β , Hypothesis 5 holds with*

$$\frac{R_a}{B_a} \leq 1 + \left(\frac{4\beta e^\gamma}{3(1 - \log 2)} + o(1)\right)(\log L)^{-2} = 1 + O_\beta((\log L)^{-2}),$$

uniformly in squarefree Y -smooth $a \leq \sqrt{Q}$. At $\beta = 6$ the displayed constant is 46.4345... In particular $B_a > 0$ for every such a . The implied constant is proportional to β , as the displayed coefficient shows; the uniformity asserted is in a , at a fixed β .

Proof. For $a \leq \sqrt{Q}$ we have $X := N/a \in [N/\sqrt{Q}, N] \subseteq [Q^{3/2}, N]$, because $N \geq Q^2$ gives $N/\sqrt{Q} \geq Q^{3/2}$. The lower endpoint contributes a negligible error. By Lemma 7(ii), $\text{thr}/a \leq Q/a$, so $U(\text{thr}/a) \leq Q/a$, $|M_{Y,Q}(\text{thr}/a)| \leq Q/a$, and $(Q/a)/((N/a)V(Y)) = Q/(NV(Y)) \ll (\log L)/Q \rightarrow 0$. For the signed estimate, uniformly in a ,

$$\frac{Q/a}{(N/a)(\log L)^{-3}} = \frac{Q}{N}(\log L)^3 \leq \frac{(\log L)^3}{Q} \rightarrow 0.$$

Hence by Lemma 16, Theorem 20 and $V(Y) = (1 + o(1))e^{-\gamma}/(\beta \log L)$ from (M3),

$$U_a \geq (1 - \log 2 + o(1))\frac{N}{a}V(Y), \quad |M_a| \leq \left(\frac{2}{3} + o(1)\right)\frac{N}{a}(\log L)^{-3},$$

$$r := \frac{|M_a|}{U_a} \leq \frac{(\frac{2}{3} + o(1))(\log L)^{-3}}{(1 - \log 2 + o(1))V(Y)} = \frac{\frac{2}{3}\beta e^\gamma + o(1)}{(1 - \log 2)(\log L)^2}.$$

So $|M_a| < U_a$ for L large, whence $B_a \geq (U_a - |M_a|)/2 > 0$, and (26) gives $R_a/B_a \leq (1+r)/(1-r) = 1 + 2r + O(r^2)$, which is the display. Every estimate depends on a only through $X = N/a$, and Lemma 16 and Theorem 20 are uniform on $[Q^{3/2}, N]$. This proves the asserted uniformity in a . $\square$

7 The normalisation

This section proves Proposition 8. Write $Y_1 = \text{thr}$, $P_1 = \sum_{n \in \mathcal{B}} \mu(n)$,

$$\Phi_\mu(x) = \sum_{m \leq x, P^+(m) \leq Q} \mu(m), \quad F(y) = \#\{n \leq y : \mu^2(n) = 1, P^+(n) \leq Q\}, \quad \Delta_\pi = \pi - \text{li}.$$

These quantities are independent of Y .

Lemma 22. For every integer $N \geq 4$,

$$P_1 = M(N) - M(Y_1)(1 + \pi(Q)) + \sum_{k \leq Y_1} \mu(k)\pi(N/k).$$

Proof. $P_1 = \Phi_\mu(N) - \Phi_\mu(Y_1)$, and $\Phi_\mu(Y_1) = M(Y_1)$ because $Y_1 = \text{thr} \leq Q$ by Lemma 7(ii) makes every $m \leq Y_1$ automatically Q -friable. For $\Phi_\mu(N)$: by Lemma 7(i) a squarefree $m \leq N$ has at most one prime factor exceeding Q , and if $m = pm'$ with $p > Q$ then $m' \leq \lfloor N/q_1 \rfloor = Y_1 \leq Q$, so m' is Q -friable, $p \nmid m'$ and $\mu(m) = -\mu(m')$. Summing, $M(N) = \Phi_\mu(N) - \sum_{Q < p \leq N} \Phi_\mu(N/p) = \Phi_\mu(N) - \sum_{Q < p \leq N} M(N/p)$. Finally $\sum_{Q < p \leq N} M(N/p) = \sum_k \mu(k) \#\{p : Q < p \leq N/k\} = \sum_{k \leq Y_1} \mu(k)[\pi(N/k) - \pi(Q)]$, the terms with $k > Y_1$ vanishing because then $N/k < q_1$ and no prime lies in (Q, q_1) ; and $\sum_{k \leq Y_1} \mu(k) = M(Y_1)$. $\square$

Lemma 23. With $T = \sum_{k \leq Y_1} \mu(k)\text{li}(N/k)$,

$$T = M(Y_1)\text{li}(N/Y_1) + N \int_1^{Y_1} \frac{M(v)}{v^2 \log(N/v)} dv.$$

Proof. Write $\text{li}(N/k) = \text{li}(2) + \text{Li}(N/k)$ with $\text{Li}(y) = \int_2^y dt/\log t$, legitimate since $N/k \geq N/Y_1 \geq q_1 > Q \geq 2$. Then

$$\sum_{k \leq Y_1} \mu(k) \int_2^{N/k} \frac{dt}{\log t} = \int_2^N \frac{M(\min(Y_1, N/t))}{\log t} dt = M(Y_1)\text{Li}(N/Y_1) + \int_{N/Y_1}^N \frac{M(N/t)}{\log t} dt,$$

the split being legitimate because $N/Y_1 \geq q_1 > 2$; and $v = N/t$ turns the last integral into $N \int_1^{Y_1} M(v)v^{-2} \log(N/v)^{-1} dv$. Adding $\text{li}(2)M(Y_1)$ and using $\text{li}(2) + \text{Li}(y) = \text{li}(y)$ gives the statement. $\square$

Lemma 24. Put $J_i = \int_1^{Y_1} M(v)(\log v)^i v^{-2} dv$ and $m_i = \sum_{k \leq Y_1} \mu(k)(\log k)^i/k$. Then

- (i) $|J_i| \leq \Lambda_i = O_i(1)$ for every $i \geq 0$, uniformly in Y_1 ;
- (ii) $J_0 = m_0 - M(Y_1)/Y_1$ and $J_1 = m_1 + J_0 - M(Y_1) \log Y_1/Y_1$;
- (iii) $m_0 = O(\eta_0)$ and $m_1 = -1 + O(\eta_1)$, where $\eta_j \ll (\log Y_1)^{(4j+3)/4} \exp(-c_M(\log Y_1)^{1/4})$.

Proof. (i) On $[1, 2)$ use $|M(v)| = |M(\lfloor v \rfloor)| \leq v$, so that

$$\int_1^2 |M(v)|(\log v)^i \frac{dv}{v^2} \leq \int_1^2 (\log v)^i \frac{dv}{v} = \frac{(\log 2)^{i+1}}{i+1}.$$

On $[2, \infty)$ apply (L) at the real argument v , which is legitimate since $M(v) = M(\lfloor v \rfloor)$ and (L) is stated for real $x \geq 2$; then substitute $w = \log v$ and extend the lower limit from $\log 2$ down to 0, so as to obtain

$$\begin{aligned} \int_2^\infty |M(v)|(\log v)^i \frac{dv}{v^2} &\leq C_M \int_{\log 2}^\infty e^{-c_M w^{1/4}} w^i dw \leq C_M \int_0^\infty e^{-c_M w^{1/4}} w^i dw \\ &= 4C_M(4i+3)! c_M^{-(4i+4)}. \end{aligned}$$

(ii) is partial summation over the unit steps of M , with Y_1 an integer: $J_0 = \sum_{k < Y_1} M(k)(1/k - 1/(k+1))$ telescopes to $m_0 - M(Y_1)/Y_1$, and with $a_k = (\log k + 1)/k$, $J_1 = \sum_{k < Y_1} M(k)(a_k - a_{k+1}) = m_1 + m_0 - M(Y_1)(1 + \log Y_1)/Y_1$, which is the stated form. (iii) By Lemma 6 the completed series are 0 and -1 , so m_j differs from its limit by $-\sum_{k > Y_1} \mu(k)(\log k)^j/k$, which partial summation and (L) bound by η_j as displayed. $\square$

Since $\log Y_1 \geq \log(Q/2 - 1) \geq L - \log 4$ for $Q \geq 8$ by Lemma 7(ii), both η_j are $\ll L^{(4j+3)/4} \exp(-c' L^{1/4})$, smaller than every power of $1/L$.

Proof of Proposition 8. By Lemma 22 and $\pi = \text{li} + \Delta_\pi$,

$$P_1 = \underbrace{M(N)}_{\mathcal{R}_1} - \underbrace{M(Y_1)(1 + \pi(Q))}_{\mathcal{R}_2} + \underbrace{\sum_{k \leq Y_1} \mu(k) \Delta_\pi(N/k)}_{\mathcal{R}_3} + T.$$

By (L), $|\mathcal{R}_1| \leq C_M N \exp(-c_M (\log N)^{1/4})$. By (L) and (C),

$$|\mathcal{R}_2| \leq C_M Q \exp(-c_M (\log Y_1)^{1/4}) (1 + 1.26 Q/L) \leq 3C_M N \exp(-c' L^{1/4})/L,$$

using $1 \leq Q/L$ and $Q^2 \leq N$. For $\mathcal{R}_3$, every argument satisfies $N/k \geq N/Y_1 \geq q_1 > Q$, so $\log(N/k) > L$ and (P) gives

$$|\mathcal{R}_3| \leq C_\pi N e^{-c_\pi \sqrt{L}} \sum_{k \leq Y_1} \frac{1}{k} \leq C_\pi N (1 + L) e^{-c_\pi \sqrt{L}}.$$

All three are $\ll N \exp(-c'' L^{1/4})$, hence $o(N/(\log N)^A)$ for every A .

For T , Lemma 23 together with the elementary estimate $\text{li}(y) = (y/\log y)(1 + O(1/\log y))$ gives

$$|M(Y_1) \text{li}(N/Y_1)| \ll \frac{N}{L} \exp(-c' (\log Y_1)^{1/4}),$$

which is again negligible. For the integral, put $w = \log v / \log N$, which lies in $[0, 1/2]$ by Lemma 7(ii)–(iii), and expand $(1 - w)^{-1} = 1 + w + w^2(1 - w)^{-1}$, the last term here being at most $2w^2$:

$$\int_1^{Y_1} \frac{M(v) dv}{v^2 \log(N/v)} = \frac{1}{\log N} \left[J_0 + \frac{J_1}{\log N} + E \right], \quad |E| \leq \frac{2\Lambda_2}{(\log N)^2}.$$

By Lemma 24(ii)–(iii), $J_0 = O(\eta_0 + C_M e^{-c' L^{1/4}})$ and $J_1 = -1 + O(\eta_1 + \eta_0 + L C_M e^{-c' L^{1/4}})$, so

$$T = -\frac{N}{(\log N)^2} + O\left(\frac{N\eta_0}{\log N}\right) + O\left(\frac{N\eta_1}{(\log N)^2}\right) + O\left(\frac{2N\Lambda_2}{(\log N)^3}\right) + O\left(\frac{N}{L} e^{-c' L^{1/4}}\right).$$

Each error is $O(N/(\log N)^3)$ once $\log N$ exceeds a threshold determined by (C_M, c_M, C_π, c_π) alone, which gives the displayed asymptotic. In particular $P_1 < 0$, and for every $\varepsilon > 0$ one has $|P_1| \geq (1 - \varepsilon)N/(\log N)^2$ for $\log N$ large; with $\log N < 2L + 2/Q$ from Lemma 7(iii) this yields $|p_1| = |P_1|/R \geq (1 - \varepsilon')(N/R)/(4L^2)$, where $R > 0$ follows from $P_1 = |\mathcal{E}| - R < 0$, and $\|p\|_2 \geq |p_1|$ because $d = 1$ is a coordinate.

For the band size, the bijection of Lemma 22 applied to the squarefree indicator gives the exact identity $F(N) = Q_2(N) - \sum_{Q < p \leq N} Q_2(N/p)$. By (15), $Q_2(x) = \frac{6}{\pi^2}x + O(\sqrt{x})$; by (P), $\sum_{Q < p \leq N} 1/p = \log(\log N / \log Q) + O(L^{-2}) = \log 2 + O(L^{-2})$; and by (C) with partial summation, $\sum_{Q < p \leq N} \sqrt{N/p} \ll N/L$. Hence $F(N) = \frac{6}{\pi^2}(1 - \log 2)N + O(N/L)$, and since $F(\text{thr}) \leq \text{thr} \leq Q$ we get $|\mathcal{B}| = \frac{6}{\pi^2}(1 - \log 2)N + O(N/L)$. Finally $|\mathcal{B}| = |\mathcal{E}| + R$ and $P_1 = |\mathcal{E}| - R$, so $R = \frac{1}{2}(|\mathcal{B}| - P_1) \geq \frac{1}{2}|\mathcal{B}|$ once $P_1 \leq 0$; as $\frac{3}{\pi^2}(1 - \log 2) = 0.0932720\dots$, this gives $R \geq 0.09N$ for L large, and $N/R \rightarrow \pi^2/(3(1 - \log 2))$ follows from $P_1 = o(N)$. $\square$

Remark 25. The identity $\sum_{n \geq 1} \mu(n)/n = 0$ removes the possible term of order $N/\log N$, while $\sum_{n \geq 1} \mu(n) \log n/n = -1$ gives the main term of P_1 . The sign $P_1 < 0$ implies $R > 0$, so the lower bound for $|p_1|$ can be obtained without the density calculation. The latter separately gives $N/R = O(1)$ and its limiting value.

In the final estimate, the factor N/R from the row bounds, including the Rankin bound for δ , cancels against the factor R/N from $1/\|p\|_2$. Thus the relative estimate uses the lower bound for $\|p\|_2$, but does not require a numerical bound for N/R .

8 Proof of Theorems 2 and 3

Let $\beta \geq 4$ be fixed, and apply Theorem 9 with an internal tolerance $\eta > 0$. By Theorem 21, Hypothesis 5 holds with $\max(1, C) = 1 + o(1)$. Insert into Theorem 9 the estimate $\sum_{p>Y} p^{-2} = (1 + o(1))/(Y \log Y) = (1 + o(1))/(\beta L^\beta \log L)$ of Lemma 12, and the bound $\delta R/N \leq \Xi \exp(-L/(2\beta \log L))$ of Lemma 14, noting that N/R cancels in the latter as well. The first term of Theorem 9 becomes

$$(4 + \eta)L^2 \sqrt{\frac{15}{\pi^2}} \cdot \frac{1 + o(1)}{\sqrt{\beta L^\beta \log L}} = (1 + o(1)) \frac{4 + \eta}{\sqrt{\beta}} \sqrt{\frac{15}{\pi^2}} \cdot \frac{L^{2-\beta/2}}{\sqrt{\log L}}, \quad (27)$$

and the second becomes

$$(4 + \eta)L^2 \left(2\Xi \exp\left(-\frac{L}{2\beta \log L}\right) \right)^{1/2} = (4 + \eta) \sqrt{2e^{e-1}} L^2 \exp\left(-\frac{L}{4\beta \log L}\right), \quad (28)$$

where $\Xi = e^{e-1}$.

Given $\varepsilon > 0$, choose $\eta > 0$ small enough that the coefficients in (27) and (28), before the $o(1)$ term, are below $4\beta^{-1/2} \sqrt{15/\pi^2} + \varepsilon/2$ and $4\sqrt{2e^{e-1}} + \varepsilon/2$, respectively. For sufficiently large N , the remaining $o(1)$ term is absorbed into the tolerance. This proves Theorem 3. $\square$

For Theorem 2, fix $A > 0$ and apply Theorem 3 with $\beta = 2A + 4$. The first term is $\ll_A L^{2-\beta/2} (\log L)^{-1/2} = L^{-A} (\log L)^{-1/2} \leq L^{-A}$, and the second is $\ll_A L^{-A}$ because $\exp(-L/((8A+16)\log L))$ decays faster than every power of L . Hence $\kappa(N) \ll_A L^{-A} = (\log Q)^{-A}$. $\square$

Corollary 4 follows by setting $\beta = 6$ in Theorem 3. For its second bound, use

$$L^2 \exp(-L/(24 \log L)) = o((L\sqrt{\log L})^{-1})$$

and $c_1 < 2.01317$. $\square$

Remark 26. For fixed $\beta \geq 4$ and $\eta > 0$, the first term in (27) is $O_{\beta,\eta}(L^{2-\beta/2}/\sqrt{\log L})$ and tends to zero, including at $\beta = 4$. The constant c_1 in Corollary 4 depends on the choice $\beta = 6$. Allowing β to grow with N would require estimates uniform in β in Lemmas 15, 17, 19, and Theorem 21; the present argument does not provide them.

9 Effectivity

All inputs are effective: (P) and (L) use no Siegel-type or zero-density estimate. The normalisation asymptotic in Proposition 8 depends on (C_M, c_M, C_π, c_π) through Lemma 24, where $\Lambda_2 = O(1) + 4C_M \cdot 11! c_M^{-12}$. The overall threshold $N_0(\beta, \varepsilon)$ in Theorem 3 also incorporates the thresholds in Theorems 9 and 21. We evaluate the leading coefficients in Corollary 4, but not these thresholds or the constant implied by $\ll_A$ in Theorem 2.

Two scale comparisons explain why the present estimates do not give useful bounds at the sizes computed in §11.

1. *The formal two-term bound becomes small before the counting estimates apply.* Ignoring the error tolerances, the expression $c_1/(L\sqrt{\log L}) + c_2 L^2 \exp(-L/(24 \log L))$ falls below 1 at $L = 3766.52 \dots$, corresponding to $\log_2 N = 10867.87 \dots$. This is not a validity threshold for the theorem. At that scale, $Z_0 = \exp(L/(\log L)^3) \approx 852$, while $Y = L^6 \approx 2.9 \cdot 10^{21}$, so $u = \log Z_0 / \log Y \approx 0.14 < 1$. Lemmas 15, 17, and 19 require $u \rightarrow \infty$. The quantity $L/(6(\log L)^4)$ reaches 1 only near $L = 1.08 \cdot 10^5$, or $\log_2 N \approx 3.1 \cdot 10^5$.
2. *The choice $Y = L^6$ gives zero weights at the evaluated sizes.* The truncation in (10) retains only classes with $a_Y \leq \sqrt{Q}$. For $\beta = 6$, the condition $Y < Q$ is $6 \log L < L$, whose large crossing is $L = 16.9988 \dots$. The corresponding continuous values of $\log_2 N$, rounded upward to two decimals, are 49.05 for $Y < Q$ and 132.51 for $Y \leq \sqrt{Q}$. At $N = 2^{28}$, the

largest size with two-sided certificates below, $Y = L^6 \approx 8.4 \cdot 10^5 > Q$. Every band element is then Y -smooth and exceeds $\sqrt{Q}$, so $x = 0$. The same holds at the other evaluated sizes when $Y = L^6$.

The construction can be nonzero even when $Y > \sqrt{Q}$. For example, at $N = 2^{52}$, $Q = 67108864$, $L = 18.0218\dots$, $Y = 34260433.5\dots < Q$, $q_1 = 67108879$, and $\text{thr} = 67108849$. The primes 34260449 and 34260493 lie in $(Y, Q]$, and $2 \cdot 34260449 \in \mathcal{E}$, $2 \cdot 34260449 \cdot 34260493 \in \mathcal{O}$ both have small-prime part 2. Thus $B_2, R_2 > 0$ and (10) assigns positive weight to this class. The finite-size construction bounds in §11 instead use much smaller choices of Y .

The estimate $|e_d| \leq \max(t_d, (Ax)_d)$ in Lemma 13 does not use cancellation between t_d and $(Ax)_d$. Controlling that cancellation may improve the constants and reduce the threshold; we do not pursue such estimates here.

10 Partial formalisation

The accompanying Lean 4 file `KappaZero.lean` uses toolchain `leanprover/lean4:v4.33.1` and `mathlib v4.33.1`. It contains no `sorry` and uses only the axioms `propext`, `Classical.choice`, and `Quot.sound`. The formalised results are the finite statements in Lemma 7(i) and the upper bound in part (ii); the equivalence $d \mid n \Leftrightarrow d \mid a_Y(n)$ on smooth rows and the row identity (18), including classes with $B_a = 0$; the M - π identity in Lemma 22; the Bonferroni identity in Lemma 15 and its exact truncation error; the parity split and ratio bound (26); and the cone membership $Ax \in K_Q$ for $x \geq 0$. The Bertrand lower bound and logarithmic inequalities in Lemma 7 are not formalised.

Several statements are proved in a general conditional form. The row identity concerns an arbitrary finite set of squarefree integers; the parity split assumes the two class-count identities; and the cone lemma assumes nonnegative coefficients. The file does not instantiate all these statements at the band (2): it contains no formal definition of $\kappa(N)$ for this band and no theorem about its asymptotic behaviour.

The analytic inputs and estimates are outside the formalisation. These include Mertens' theorems, (P), (L), the Rankin, sieve, friable, and rough-Möbius estimates, and Proposition 8. Consequently Theorems 9, 20, 21, and 2 are not formally verified.

11 Certified finite-size bounds

Because K_Q is finitely generated, $\text{dist}_2(c_{\mathcal{O}}, K_Q)$ is the value of a nonnegative least-squares problem whose data are integral, and its square is rational. Let π be the projection of $c_{\mathcal{O}}$ on K_Q , which exists and is unique because K_Q is a closed convex cone. If $\pi = 0$ then $\text{dist}_2(c_{\mathcal{O}}, K_Q)^2 = \|c_{\mathcal{O}}\|_2^2$ is an integer. Otherwise write $\pi = \sum_{\ell} y_{\ell} A_{\ell}$ with $y \geq 0$, which is possible because $\pi \in K_Q$, and among all such representations take one whose support $S = \{\ell : y_{\ell} > 0\}$ has fewest elements; there are finitely many generators, so such a representation exists, and $S \neq \emptyset$. Its columns are then linearly independent. For suppose $\sum_{\ell \in S} v_{\ell} A_{\ell} = 0$ with $v \neq 0$ supported on S ; replacing v by $-v$ if necessary, some $v_{\ell} > 0$. Then $y - tv$ represents π for every t , and it stays nonnegative for $0 \leq t \leq t^* := \min\{y_{\ell}/v_{\ell} : v_{\ell} > 0\} > 0$, since the coordinates with $v_{\ell} \leq 0$ only increase. At $t = t^*$ a further coordinate vanishes, giving a nonnegative representation of π with strictly smaller support and contradicting the choice of S . Let B collect those independent columns and y_S the corresponding coefficients. Because $y_{\ell} > 0$ on S , the optimality conditions for the projection give $B^{\top}(c_{\mathcal{O}} - \pi) = 0$, that is the normal equations $B^{\top}B y_S = B^{\top}c_{\mathcal{O}}$ with $B^{\top}B$ invertible, so $y_S = (B^{\top}B)^{-1}B^{\top}c_{\mathcal{O}}$ and $\pi = B y_S$ are rational because B and $c_{\mathcal{O}}$ are. Hence $\text{dist}_2(c_{\mathcal{O}}, K_Q)^2 = \|c_{\mathcal{O}} - \pi\|_2^2$ is rational, and so is $\kappa(N)^2$ for every N for which $\kappa(N)$ is defined, and both endpoints of an interval for $\kappa(N)$ can in principle be certified in exact

arithmetic. We give two-sided certificates at $N = 2^{16}, 2^{18}, \dots, 2^{28}$ and a lower certificate at $N = 2^{30}$. Appendix A describes the verification procedure, its format and index checks, its overflow guards, and its cost.

A pair of certificates suffices. Write $P = c_{\mathcal{E}} - c_{\mathcal{O}}$, so that $\kappa(N) = \text{dist}_2(c_{\mathcal{O}}, K_Q) / \|P\|_2$ by (7). Let $q \geq 0$ be an integral vector and $D \geq 1$ an integer (a rational primal q/D), and let λ be an integral dual with $\theta_{\lambda}(\ell) \geq 0$ for every $\ell \in \mathcal{E}$ and $\langle c_{\mathcal{O}}, \lambda \rangle < 0$. Then

$$\frac{\langle c_{\mathcal{O}}, \lambda \rangle^2}{\|P\|_2^2 \|\lambda\|_2^2} \leq \kappa(N)^2 \leq \frac{\|D c_{\mathcal{O}} - Aq\|_2^2}{D^2 \|P\|_2^2}, \quad (29)$$

and both bounds are rational and checkable in integer arithmetic from the two stored vectors. The left inequality is Proposition 1 applied to $\lambda/\|\lambda\|_2$; the right is $\text{dist}_2(c_{\mathcal{O}}, K_Q) \leq \|c_{\mathcal{O}} - Aq/D\|_2$. The dual feasibility has to be verified on all of $\mathcal{E}$, not only on the columns a solver returns; that check is a single pass over the band, and it was run at every size below.

Candidate generation and feasibility. Floating-point optimisation supplies candidate primal and dual vectors. Rounding and clamping the primal gives an integral vector $q \geq 0$. The dual is scaled, rounded, and, if necessary, repaired at $d = 1$: adding

$$\rho = \max(0, -\min_{\ell \in \mathcal{E}} \theta_{\lambda}(\ell))$$

to λ_1 makes every even-column divisor sum nonnegative, since $1 \mid \ell$ for every column. The repaired vector must still satisfy $\langle c_{\mathcal{O}}, \lambda \rangle < 0$. Appendix A gives the scaling rules and the effect of the repair.

The certified intervals. For the seven two-sided instances below, both bounds in (29) were evaluated in exact integer arithmetic. The endpoints below are the square roots of the two rational bounds that (29) gives for κ^2 , rounded outward at twelve decimals; the last column repeats them rounded outward at seven.

e	certified $\kappa(2^e) \geq$	certified $\kappa(2^e) \leq$	outward at 7 decimals
16	0.172389351665	0.172389351666	$0.1723893 < \kappa < 0.1723894$
18	0.140860453831	0.140860453832	$0.1408604 < \kappa < 0.1408605$
20	0.105542708383	0.105542708385	$0.1055427 < \kappa < 0.1055428$
22	0.076331197680	0.076331197683	$0.0763311 < \kappa < 0.0763312$
24	0.052762952627	0.052762952635	$0.0527629 < \kappa < 0.0527630$
26	0.034622922800	0.034623479647	$0.0346229 < \kappa < 0.0346235$
28	0.024526262835	0.024538226032	$0.0245262 < \kappa < 0.0245383$

The widths are 1.0, 1.0, 2.0, 3.0, 8.0 in units of 10^{-12} at $e = 16, \dots, 24$, at most $5.6 \cdot 10^{-7}$ at $e = 26$, and at most $1.2 \cdot 10^{-5}$ at $e = 28$. At $e = 16, 18, 20$, dense nonnegative least squares used all distinct even-parity columns: 4414, 18685, and 76946, respectively. At $e = 22, \dots, 28$, column generation used a restricted master problem, with a sweep over the full band for pricing at each round. Restricting the column set can only increase the distance, so a rounded nonnegative solution still gives a valid upper certificate. At $e = 26, 28$, the stored primal and dual were selected from different runs, using the best available upper and lower bounds. The intervals refer to these fixed stored vectors, each verified independently.

$e = 30$: the lower endpoint only. At $N = 2^{30}$ the band has $|\mathcal{E}| = 86\,592\,333$ and $|\mathcal{O}| = 89\,520\,933$, and only the left half of (29) has been evaluated. Writing

$$J = -\langle c_{\mathcal{O}}, \lambda \rangle, \quad H = \|\lambda\|_2^2, \quad D_p = \|P\|_2^2, \quad (30)$$

so that the left half of (29) reads $\kappa(N)^2 \geq J^2/(HD_p)$ when $J > 0$, the stored run records the four integers J , H , D_p and c_6 with

$$c_6^2 H D_p < J^2 \cdot 10^{12}, \quad (31)$$

which is that half cleared of denominators and certifies $\kappa(N) > c_6/10^6$. Here $J = 4401612119558212802 > 0$, $H = 2015370564893177546709684150$, $D_p = 30382600959846$ and $c_6 = 17787$, the largest six-decimal value for which (31) holds, so

$$\kappa(2^{30}) > 0.017787.$$

This bound also requires the stored integral λ and the full-band feasibility check: (31) alone does not certify it. The archived data include the repaired integral vector and the four integers above. No integral primal was produced at 2^{30} , so no upper endpoint is claimed there; the restricted master reports $\kappa(2^{30}) \approx 0.01779$, which is a solver value and not certified.

The construction's own bounds. The construction (10), evaluated with Y chosen independently of L^6 as discussed in §9, gives $\kappa \leq 0.6982, 0.6130, 0.6088, 0.5841$ at $e = 16, 18, 20, 22$, attained at $Y = 3, 3, 5, 5$. Each squared bound is an exact rational computed from the integer class counts R_a, B_a through (10). The bounds themselves are irrational at these four sizes; the displayed values are rounded upward to four decimals. They exceed the certified optimum by factors between four and eight, and over these four sizes the ratio to the optimum increases with N .

The computed densities are still below the asymptotic range in Proposition 8. Across the eight sizes, R/N increases from 0.0768 at 2^{16} to 0.0834 at 2^{30} , remaining below both 0.09 and the limit 0.0932720.... Correspondingly, N/R decreases from 13.03 to 11.99, above its limit 10.7213228.... This is consistent with the proposition, whose inequalities hold only for sufficiently large N .

A Verification of the certificates

The archived software and data are identified in the data availability statement; the archive includes the verification commands and identifies the runs that produced the stored outputs.

Candidate generation. For a floating primal y , choose an integer $D \geq 1$ for each instance and set $q = \max(0, \lfloor Dy \rfloor)$ componentwise. At the five smaller sizes the dual candidate is obtained from the solver residual $r = c_{\mathcal{O}} - Ay$: it is $-r$, scaled by $K/\max_d |r_d|$ and rounded componentwise, where K is an integer. Before repair this gives $\max_d |\lambda_d| \leq K$. At $e = 26, 28$, no residual is formed. The solver instead stores a floating dual w with $\langle c_{\mathcal{O}}, w \rangle > 0$, and $-w$ is scaled and rounded in the same way.

Before being stored, the floating dual at these two sizes is repaired row-wise. A column whose divisor sum exceeds a threshold is assigned to its divisor row of least t_d , excluding $d = 1$. The coordinate in that row is decreased by the largest amount required by its assigned columns. The threshold producing the best floating lower bound is retained. The primal is selected from the round with the best upper bound and the dual from the round with the best lower bound; they need not come from the same round or run.

After rounding, dual feasibility is repaired at $d = 1$ by the amount ρ defined in §11. This changes $\langle c_{\mathcal{O}}, \lambda \rangle$ by ρR and $\|\lambda\|_2^2$ by $2\rho\lambda_1 + \rho^2$, where λ_1 here denotes the pre-repair coordinate. The norm increases strictly if and only if $\rho > 0$ and $\lambda_1 > -\rho/2$. The latter condition holds for all stored certificates, but the repair may be zero: at $e = 20$ the rounded dual already has strictly positive minimum divisor sum. The comparison with a stored floating optimum is only a diagnostic, not a general consequence of repair. At the five sizes with this diagnostic, the

certified lower endpoint does not exceed the recorded floating optimum and is strictly below it when $\rho > 0$. The stored vector is the repaired integral vector, not the original floating candidate.

Exact verification. The command `pa_certify.py --verify` checks the seven stored two-sided certificates. The two largest were constructed by `pa_certify_big.py`, which also verifies them. From the exponent e alone, the verifier derives $N = 2^e$, $Q = \lfloor \sqrt{N} \rfloor$, q_1 , thr , and the complete list of squarefree rows $d \leq Q$ from a Möbius sieve. The cached instance and certificate must agree with this derivation. The verifier then reconstructs the band, compares $|\mathcal{E}|$, $|\mathcal{O}|$, $c_{\mathcal{E}}$, and $c_{\mathcal{O}}$ with the cached counts, checks $\theta_{\lambda} \geq 0$ on every even-parity column, confirms $\langle c_{\mathcal{O}}, \lambda \rangle < 0$ and $q \geq 0$, and recomputes both rational bounds in (29). A missing certificate fails the inventory check rather than reducing the set of reported instances.

Every index is validated before use and independently of the cached band. Row indices must be strictly increasing, squarefree, and in $[1, Q]$; column indices must be distinct, squarefree, in $(\text{thr}, N]$, have largest prime factor at most Q , and have even ω . Stored arrays must have integer types and retain all values when converted to signed 64-bit integers. Fractional entries and value-changing conversions are rejected. The two integer kernels are guarded by the exact sums of the input vectors, and their totals are checked with arbitrary-precision integer arithmetic. The scales K and D are chosen to satisfy these guards. Cached counts are compared with the reconstructed band rather than trusted.

Verification reads and checks stored certificates; construction regenerates and rewrites them. The scripts distinguish these modes, and only verification checks the archived instances.

Verification cost. One recorded run checks all seven certificates in 85.7 seconds using four threads on the author’s machine: 7.8 seconds for $e \leq 24$ together, 14.1 seconds at $e = 26$, and 63.8 seconds at $e = 28$, including reconstruction of the band. Checking the 2^{30} lower certificate takes a few additional minutes, again dominated by band reconstruction. The archived timing record specifies the hardware and commands. The reported run starts with an empty compiler cache and includes the initial kernel compilation; timings from two warm-cache runs are recorded alongside it. The machine was running unrelated work during all runs, and the largest-instance reconstruction time varies substantially. These measurements indicate the cost on that machine and are not a reproducible timing guarantee. The timing record is not overwritten by verification.

Diagnostics. Two recorded comparisons are separate from certificate validity. The first compares the stored floating optimum with the certified interval at the five smaller sizes. At $e = 16, 18$, the stored nine-decimal ledger value lies outside the twelve-decimal interval because of rounding; at the next three sizes the float is stored with the dual. No floating optimum is compared at $e = 26, 28$. The second comparison tests the internal target width 10^{-6} for the two larger certificates. It is met at 2^{26} but not at 2^{28} , whose width is nevertheless below the $1.2 \cdot 10^{-5}$ stated in §11. Neither comparison changes validity: the exact interval is determined by the stored primal and dual, and the scripts’ exit codes report certificate validity alone.

The lower certificate at 2^{30} . Here the floating dual w , with $\langle c_{\mathcal{O}}, w \rangle > 0$, is rounded at the fixed scale $\lambda = \lfloor -10^{12}w \rfloor$. The coordinate at $d = 1$ is then set to zero and replaced by the repair amount ρ . Thus the stored λ_1 equals ρ . This fixed scale reproduces the integers J and H in §11; it is not chosen to normalise $\max_d |\lambda_d|$ to an integer K , as at the smaller sizes. The sum $\sum_d |\lambda_d|$ is checked against the accumulator bound 2^{62} at the chosen scale. Apart from rounding and the rebuilt first coordinate, the stored vector is $-10^{12}w$.

In this constructor, the post-repair sweep uses the accumulator bound already checked before repair. To justify this, write A for the sum of the positive coordinates and B for the sum of the absolute values of the negative coordinates before repair, when $\lambda_1 = 0$. The guarded sum

is $A + B$. Every divisor sum is a signed subset sum of the coordinates and is at least $-B$, so $0 \leq \rho \leq B$. After repair, every partial sum in the kernel lies between $\rho - B$ and $\rho + A$, hence between $-B$ and $A + B$. Its absolute value is therefore bounded by the previously guarded $A + B$. The repaired vector must also pass the full guard before the certificate is stored. The verifier, in contrast to this constructor, checks the guard before every kernel call.

Verification uses only the stored integral vector. It derives the instance, reconstructs the band, checks every even-parity column, recomputes J, H, D_p , and confirms that c_6 is maximal. The same integer-format requirements and accumulator guards apply, and no solver is used. Every validity decision is an integer or rational comparison. Floating-point arithmetic is confined to loading machinery, printed diagnostics, and one square root used as a sieve loop bound. The verifier checks that the truncated square root agrees with the exact integer square root throughout the range where the sieve uses it.

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Data availability statement. The verification code, certificates, computational outputs, and Lean 4 development are archived at the fixed tag **P7-v1** in the following repository, under the directory listed below:

<https://github.com/soke91/arithmetic-convolution-fields>
P7-parity-separation-tends-to-zero/